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SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

A CAPSTONE PROJECT REPORT INDUSTRIAL EMPLOYEE SAFETY AND OCCUPATIONAL HEALTH MANAGEMENT SYSTEM

Submitted in partial fulfilment of the requirements for the course
DSA0101 – OOPS with C++ | Slot D | Capstone Project – Review 2

Bachelor of Technology / Bachelor of Engineering
Electronics and Communication Engineering

Submitted by

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Under the Supervision of

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Industrial Employee Safety and Occupational Health Management System” has been carried out by Dharshidha S (1911260095) under the supervision of Dr. K. Parimala Kanaga Devan, Professor, SIMATS Engineering, and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Electronics and Communication Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I Deepika S, of the Electronics and Communication Engineering Department, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Industrial Employee Safety and Occupational Health Management System” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been followed throughout the planning, implementation, analysis, and documentation of the project.

Place: Chennai

Date: September 2026

Signatures of the Students:
Dharshidha S

INDIVIDUAL CONTRIBUTION STATEMENT

Student / Register No.	Specific Responsibilities	Design & Development	Testing & Analysis	Report Contribution	Approx. Contribution
Dharshidha S / 1911260095	Hazard monitoring and alerts	Hazard-zone and alert functionality	Hazard/event testing	Ch. 3–4 and system workflow	100%
Total					100%

ABSTRACT

Industrial workplaces such as factories, warehouses, and construction sites face continuous risks from unsafe conditions, hazardous gases, high temperatures, machinery, and other occupational hazards. Manual safety inspection and paper-based incident reporting can delay hazard identification, corrective action, and emergency response. Employee health history, personal protective equipment (PPE) compliance, and training records may also be maintained separately, making integrated safety management difficult. This project proposes an Industrial Employee Safety and Occupational Health Management System that digitizes safety compliance, manages employee and health records, monitors hazard zones, and supports incident reporting and alerts.

The proposed system is developed using Object-Oriented Programming concepts in C++. Employees, hazard zones, incidents, PPE items, and related records are represented through reusable and maintainable classes. The system provides employee management, periodic health-record management, training and certification tracking, hazard-zone classification, sensor-based hazard-event logging, incident and near-miss reporting, PPE compliance tracking, corrective-action recording, and alert generation. Role-based access is considered for employees, safety officers, and supervisors. File-based storage is used for persistence of records.

The system provides a unified digital approach to safety management and reduces dependence on fragmented paper records. It supports faster access to safety information, improved incident documentation, compliance monitoring, and emergency notification. The modular design also provides a foundation for future integration with IoT sensors, mobile applications, cloud storage, dashboard analytics, GPS-based employee tracking, and machine-learning-based predictive safety analysis. Overall, the proposed solution aims to improve workplace safety, regulatory compliance, accountability, and emergency response.

Keywords: Industrial Safety, Occupational Health, C++, Object-Oriented Programming, PPE Compliance, Hazard Monitoring

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LIST OF ABBREVIATIONS / SYMBOLS

Abbreviation	Description	Unit
OOP	Object-Oriented Programming	—
PPE	Personal Protective Equipment	—
OHS	Occupational Health and Safety	—
IoT	Internet of Things	—
GPS	Global Positioning System	—
SMS	Short Message Service	—

CHAPTER 1: INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Industrial workplaces require systematic management of employee safety, occupational health, hazardous areas, personal protective equipment, training, and incident response. The project focuses on replacing fragmented and paper-based safety processes with a unified software-driven management system. The source project identifies factories, warehouses, and construction sites as environments where accidents and health hazards can occur because of unsafe conditions, hazardous gases, machinery, and other risks.

1.2 Need for the Project

Manual inspections and paper-based incident reporting can be slow and error-prone. Employee health records, PPE compliance, and training information may exist in separate registers, making it difficult for safety personnel to obtain a complete view of employee safety status. Hazardous zones may also lack continuous monitoring, while emergency alerts can depend on manual observation and reporting. The proposed system addresses these limitations by centralizing safety-related records and providing structured hazard, incident, PPE, and health management.

1.3 Problem Statement

Manual, paper-based safety inspection and incident reporting can delay hazard detection and corrective action. Employee health, PPE compliance, and training records are fragmented across different registers, while hazardous zones are not continuously monitored in the proposed baseline environment. Consequently, unsafe conditions may remain unnoticed and emergency response may be delayed. A unified software-driven system is therefore required to manage employees, health records, hazard zones, incidents, PPE compliance, training information, and safety notifications within one platform.

1.4 Project Aim

To develop a unified Industrial Employee Safety and Occupational Health Management System using C++ and Object-Oriented Programming principles to digitize employee safety records, hazard monitoring, incident reporting, PPE compliance, and emergency notification.

1.5 Project Objectives

Develop an Industrial Employee Safety and Occupational Health Management System using C++.

Apply classes, objects, encapsulation, inheritance, polymorphism, abstraction, constructors, and file handling to model safety entities.

Store and manage employee profiles, health records, training and certification status.

Classify hazard zones by risk level and log sensor-based hazard events.

Digitize incident and near-miss reporting with severity classification and corrective-action tracking.

Track PPE issuance and compliance status for employees.

Generate alerts and notifications for supervisors during hazard or incident events.

Improve workplace safety management, compliance tracking, accountability, and emergency response.

1.6 Scope

The scope includes employee management, health and medical records, training and certification tracking, hazard-zone classification, hazard-event logging, incident and near-miss reporting, PPE issuance and compliance, corrective-action tracking, alerts, reporting, role-based access, and file-based data persistence. The current implementation does not claim full physical IoT deployment, cloud deployment, GPS tracking, mobile application support, or machine-learning prediction; these are identified as future extensions. The system assumes that valid employee, hazard, incident, PPE, and training information is entered or received through the defined interfaces.

1.7 Expected Engineering Outcome

The expected outcome is a modular C++ software system that provides a unified safety-management process. It is intended to reduce dependence on paper records, improve access to employee and safety information, support hazard and incident management, and provide a foundation for future real-time sensor and analytics integration.

CHAPTER 2: LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The project review considered recent work related to industrial worker safety, Internet of Things (IoT) monitoring, wearable sensing, real-time hazard detection, location-based safety warnings, and occupational health. The project presentation identifies literature from 2023 to 2025 and summarizes the contribution of each approach.

2.2 Review of Existing Work

Recent IoT-based research has explored real-time worker safety monitoring using wearable sensors, environmental sensors, connectivity-aware monitoring, and alert systems. The reviewed work includes systems for fall detection and location tracking, monitoring of SpO2, heart rate, gas and temperature, AI/ML-supported occupational safety analysis, environmental and wearable sensor networks, ESP32-based worker monitoring, and vibration-based monitoring for machine operators.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance
Manual / paper-based safety management	Simple to start; familiar process	Slow, fragmented, error-prone; reactive	Problem baseline
IoT worker monitoring	Real-time sensing and alerts	May focus mainly on hardware and require backend integration	Supports future sensor layer
Wearable safety systems	Can monitor worker health and location	Device/connectivity dependence	Future extension
Proposed C++ management system	Unified employee, hazard, incident and PPE management; modular OOP	Current version depends on entered/file-based data and does not itself provide full physical sensor deployment	Core software solution

2.4 Research / Engineering Gap

The project identifies a gap between sensor-focused safety solutions and an integrated, extensible software backend that combines employee records, health information, hazard zones, incidents, PPE compliance, training, alerts, and role-based access. The proposed system addresses this integration gap at the software-management level.

2.5 Proposed Contribution

The proposed contribution is a modular C++ OOP-based safety-management platform that brings multiple safety functions together. It provides reusable classes for safety entities and a structured workflow from input validation and risk/severity evaluation to alerting, record storage, and report generation.

CHAPTER 3: ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder		Requirement
Employee		View relevant personal safety, health, PPE and training information and report incidents/near-misses.
Safety Officer		Manage employee safety records, monitor hazard zones, verify PPE compliance and review incidents.
Supervisor		Receive alerts, view reports, review incidents and approve corrective actions.
Requirement Type	Requirement	Measurable Target
Functional	Employee record management	Add, update, search and view employee records
Functional	Health and training management	Store checkup and certification status
Functional	Hazard monitoring	Classify zones and log hazard events
Functional	Incident management	Record cause, severity and corrective action
Functional	PPE compliance	Record issued item and compliance status
Functional	Alerts	Raise notification when defined hazard/incident conditions occur
Non-Functional	Maintainability	Modular OOP classes
Non-Functional	Persistence	Store/retrieve records using file handling
Non-Functional	Security	Role-based access concept for defined user roles

3.2 Constraints & Applicable Standards

The source project specifies functional safety and compliance requirements but does not identify particular external standards or codes as having been implemented in the system. Therefore, no specific standard is claimed as implemented in this report. Applicable regulatory and institutional standards should be identified and validated with the project supervisor before production deployment.

Constraint / Area	Requirement	Application
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Technical	C++ console-based implementation and file persistence	C++ OOP classes and .dat storage
Safety	Avoid unsafe dependence on manual reporting alone	Hazard and incident records with alert workflow
Ethical	Protect employee information and maintain accountability	Role-based access concept and controlled records
Operational	Usable by different safety roles	Employee, Safety Officer and Supervisor roles

3.3 Design Specifications

Parameter	Target / Requirement	Priority
Programming language	C++	High
Architecture	Layered, modular OOP design	High
Storage	File-based .dat persistence	High
Users	Employee / Safety Officer / Supervisor	High
Hazard data	Zone risk level and event log	High
Safety records	Health, training, PPE and incidents	High
Notifications	Hazard/incident alerts	High

3.4 Alternative Solutions & Evaluation

Alternative 1: Continue with manual registers and paper-based reporting. This requires minimal software development but does not solve fragmentation, delayed reporting, or centralized record access.

Alternative 2: Use a sensor-first IoT safety platform. This can provide real-time physical monitoring but introduces hardware, connectivity, integration, and deployment complexity.

Alternative 3: Implement a unified C++ software management system with future IoT integration. This provides a practical software foundation while keeping the architecture extensible for future sensing and analytics.

Criterion	Weight	Manual	IoT-first	C++ Unified System
Performance	25%	Low	High	Medium–High
Cost	15%	High	Low–Medium	High
Safety	25%	Low	High	Medium–High
Sustainability	15%	Low	Medium	High
Reliability	20%	Low–Medium	Medium–High	High

3.5 Selected Design & Detailed Design

The C++ unified software system is selected because it directly addresses the project's identified integration gap while remaining modular and maintainable. Its OOP structure supports future expansion without requiring the core safety records to be redesigned. The system consists of a presentation layer, application layer containing C++ OOP classes, a data layer using file-based storage, and external notification/reporting interfaces.

System flow: User/Sensor Input → Data Validation & Classification → Risk/Severity Evaluation → Alert & Notification → Record & Report Generation → Continuous Re-check of Hazard Zones and PPE Compliance.

The Employee Management, Hazard Zone, PPE/Compliance, Health/Medical, Incident/Accident, and Alert/Notification modules exchange structured information. Employee and safety records are persisted in .dat files, while alerts and reports use the processed event information.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
File-based storage	Cloud database	Simple local deployment and suitable for academic prototype	Limited scalability and multi-site access	Selected for current scope; cloud is future work
C++ OOP backend	Monolithic procedural code	Encapsulation, reuse and maintainability	Requires OOP design effort	Selected because project is based on OOP with C++
Future IoT integration	Immediate hardware deployment	Allows gradual expansion	Current prototype lacks physical sensor deployment	Deferred to future scope

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Decision
Sustainability	Reduce paper-based records	Unified digital record approach	Supports digital documentation
Ethics	Responsible employee-data handling	Health and personal records are sensitive operational data	Role-based access is included in architecture

Health & Safety		Fast hazard and incident response	Hazard classification, alerts and corrective actions		Safety modules are core functions
Security		Controlled access to records	Employee/Safety Officer/Supervisor roles		Authentication, authorization and RBAC are included in architecture
Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect data entry	Medium	Medium	Medium	Validation and structured input	Low–Medium
Missed hazard event	Medium	High	High	Event logging and alert workflow; future real sensors	Medium
Unauthorized record access	Medium	High	High	Role-based access and authentication	Medium
File corruption/loss	Low–Medium	High	Medium–High	Backups and future cloud storage	Medium
Delayed manual response	Medium	High	High	Automated alert workflow	Medium

CHAPTER 4: IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The development approach follows modular Object-Oriented Programming. Safety entities are represented using classes and their responsibilities are separated into employee/health, hazard/alert, and incident/PPE/compliance modules. File handling provides persistence for employee, incident, and compliance records. The workflow validates and classifies input before recording events and generating appropriate alerts or reports.

Resource	Purpose
C++ compiler / GCC	Compile and execute the C++ application
Code::Blocks / Dev-C++	Development environment
Windows Operating System	Development and execution platform
.dat files	Persistent record storage
C++ OOP classes	Model system entities and operations

4.2 Software Implementation & Modern Tools Used

Tool / Technology	Purpose	Stage Used
C++	Core implementation	All development
GCC Compiler	Compilation	Testing/build
Code::Blocks / Dev-C++	Code development	Implementation
File Handling	Persistence	Implementation
OOP	System architecture	Design and implementation

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion	Required Value	Achieved Value	Status
Add employee	Create employee record and retrieve it	Record stored and retrievable	Valid employee record	Functional workflow defined	Pass
Update employee	Modify employee details	Updated details displayed	Updated values	Functional workflow defined	Pass
Health record	Store checkup information	Record available for employee	Valid health record	Functional workflow defined	Pass
Hazard event	Log gas/temperature event	Event classified and alert raised when threshold is breached	Valid event	Alert workflow defined	Pass

Incident reporting	Enter incident cause/severity	Incident saved with corrective action	Valid incident	Functional workflow defined	Pass
PPE compliance	Issue PPE and verify status	Compliance status recorded	Valid PPE record	Functional workflow defined	Pass
Training/certification	Enter certification and expiry	Status searchable	Valid certification	Functional workflow defined	Pass

4.4 Results

The project presentation reports that the proposed system provides a unified digital approach to employee, health, hazard, incident, PPE, compliance, and training management. The design supports real-time alert generation when defined hazard thresholds are breached and provides structured incident and compliance records. The presentation does not provide measured numerical performance results such as response time, accuracy percentage, throughput, or benchmark execution time; therefore, such values are not claimed in this report.

4.5 Data Analysis & Engineering Judgment

The engineering judgment is based on functional coverage and architectural suitability rather than numerical benchmarking. The modular design reduces fragmentation by bringing safety-related records into one application. OOP features provide separation of responsibilities and extensibility. A major limitation is that the current project is primarily a software management solution and does not demonstrate full physical sensor deployment; therefore, claims of actual real-world sensor response time are outside the available project evidence.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Develop C++ safety system	C++ OOP architecture and modules	Fully
Apply OOP concepts	Classes, encapsulation, inheritance, polymorphism, abstraction and constructors	Fully
Manage employee/health/training	Employee & Health Management module	Fully
Classify hazard zones and log events	Hazard Zone Monitoring module	Fully
Digitize incidents and corrective actions	Incident & Compliance module	Fully

Track PPE compliance	PPE & Compliance module	Fully
Generate alerts	Alert & Notification module	Fully
Improve workplace safety/compliance	Unified workflow and centralized records	Partially; effectiveness requires field deployment and measurement

4.7 Strengths & Limitations

Strengths: Unified safety records; modular OOP architecture; hazard and incident workflow; PPE and training tracking; role-based access concept; file-based persistence; future extensibility.

Limitations: Current implementation is based on software/file management rather than a complete deployed IoT infrastructure; numerical performance benchmarks are not provided; cloud, mobile, GPS, machine learning, and full dashboard analytics are future extensions; production deployment would require detailed security, backup, standards, and regulatory validation.

4.8 Project Planning, Roles & Budget

Milestone		Planned Activity
1		Problem identification and requirements
2		Literature survey and existing-system analysis
3		System architecture and module design
4		C++ OOP implementation
5		Testing and verification
6		Documentation and final review
Student	Assigned Responsibility	Actual Contribution
Deepika S	Employee and health management	Employee, health and training modules
Dharshidha S	Hazard monitoring and alerts	Hazard-zone and alert workflow
Monisha S	Incident, PPE and compliance	Incident/PPE/compliance and documentation
Item	Estimated Cost	Actual Cost
Development software/tools	Existing/free academic tools	Existing/free academic tools
Computer/system resources	Existing resource	Existing resource
Prototype IoT hardware	Not included in current implementation	Not incurred
Total	No additional project-specific hardware cost identified	No additional project-specific hardware cost identified

CHAPTER 5: CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The Industrial Employee Safety and Occupational Health Management System addresses the problem of fragmented, manual safety management by providing a unified C++-based software solution. The system integrates employee records, health and training information, hazard-zone management, incident reporting, PPE compliance, corrective actions, alerts, and role-based access. The modular OOP architecture supports maintainability and future extension. The project achieves its core software-management objectives, while real-world effectiveness and numerical performance require further field testing and physical deployment.

5.2 Future Work

Integrate real-time IoT sensors for gas, temperature and wearable health monitoring.

Develop machine-learning-based predictive analytics to anticipate workplace hazards.

Create a mobile application for supervisors and employees to receive alerts remotely.

Introduce cloud-based storage for centralized multi-site safety management.

Develop dashboard-based visual analytics for hazard trends and compliance tracking.

Add GPS/location-based employee tracking for large industrial sites.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

Object-Oriented Programming applied to a real engineering safety problem.

Design of modular software components and role-based workflows.

File handling for persistent safety records.

Understanding of hazard classification, incident severity and PPE compliance management.

Engineering & Professional Skills Developed

Problem formulation and requirements analysis.

System architecture and modular design.

C++ implementation and debugging.

Testing, verification, technical documentation and teamwork.

Individual Reflection

Deepika S: The most difficult part was organizing employee, health and training information into a coherent software structure. Applying classes and encapsulation helped separate data from operations. The work improved my understanding of designing reusable C++ components. If the project were repeated, I would plan stronger validation and database integration from the beginning.

Dharshidha S: Designing the hazard-zone and alert workflow required considering risk levels, event logging, and notification. The key engineering decision was to keep hazard monitoring modular so that future sensor inputs can be connected without changing the whole application. I gained practical understanding of safety-event processing. I would add more automated test cases if repeating the project.

Monisha S: Managing incident, PPE, and compliance records highlighted the importance of accurate and traceable safety documentation. File handling and abstraction helped create a maintainable structure for these records. I developed stronger skills in C++ OOP, documentation, and system-level thinking. If repeated, I would improve backup, security, and reporting features.

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APPENDICES

Appendix A – Detailed Calculations

No numerical engineering calculations were provided in the project presentation. Future quantitative validation may include alert-response time, record retrieval time, data-processing time, and system availability.

Appendix B – Engineering Drawings / Schematics

The proposed architecture consists of Users → Security Layer → Application Layer (C++ OOP Classes) → Data Layer (.dat files), with external alert and reporting components. The main modules are Employee Management, Hazard Zone Management, Health/Medical Records, Incident Management, PPE Management, and Reports/Alerts.

Appendix C – Source Code / Code Repository Information

The project is implemented in C++ using OOP concepts and file handling. The complete source code should be maintained in the approved project repository; only essential excerpts should be inserted here if required by the department.

Appendix D – Datasheets

No physical sensor datasheets are applicable to the current software-focused implementation. Datasheets should be added when IoT sensors are integrated.

Appendix E – Experimental / Raw Data

The supplied project material does not contain numerical experimental datasets. Future testing should record test inputs, expected outputs, achieved outputs, response times, and pass/fail status.

Appendix F – Ethics / Institutional Approval

The project handles employee-related health and safety information conceptually. Any real deployment using personal or medical information should follow applicable institutional, privacy, security, and ethical requirements.

Appendix G – Project Meeting / Progress Records

Maintain supervisor meeting dates, review comments, action items, and completion status here as required.

Appendix H – User / Industry Feedback

No industry feedback was included in the supplied project material. Add verified feedback if collected.

Appendix I – Publications / Patents / Competitions / Product Outcomes

No publication, patent, competition result, or product outcome was stated in the supplied material.

Appendix J – Individual Contribution Evidence

Student	Contribution Evidence
Dharshidha S	Hazard-zone monitoring, alert workflow, testing and documentation